

PD39 Physical Blocker Validation

First Compatibility Atlas - STOP Report

PD39 research branch: `research/pd39-physical-blockers-atlas-v1`

14 September 2026

FINAL CASE: C - STOP

No physically confirmed blocker; no compatibility atlas was run.

Executive verdict

The prior fresh-holdout census supplied three low-order C12 portfolios that looked like H0 candidates on the default PowerDynamics initialization path: 35;36, 37;38, and 30;32;33. This campaign tested those cases, their immediate predecessors, cheap proper subsets, and aggregate-matched controls. The default-path positive alphas were not reproducible under the frozen Gate A tolerance/path audit. All three selected C12 blocker TDS runs returned `MaxIters`; predecessors and controls completed and decayed.

Thus the candidates are not physically validated dynamic instabilities. The correct result is a model/equilibrium-path warning. The conditional atlas, composition claim, contextual-return validation, remediation, and excluded campaigns were not run.

1 Scope, provenance, and frozen decision

The final branch is `research/pd39-physical-blockers-atlas-v1`, created from parent commit `e81d18ab945b`. Preregistration was committed before new numerical results at `1b9ad5fe`. Gate A, default-path audit, and Gate B commits were `02381549`, `d24f0ad3`, and `c95bab91`. PowerDynamics was version 5.0.0 under Julia 1.11.9. No push was performed.

Only the following gates were allowed: numerical/DAE physicality, common nonlinear TDS, conditional aggregate composition, and exact network-closure feasibility. Weak-node/link ranking, structured-radius search, repair optimization, planners, co-design, IEEE-68, EMT, and new controller search were explicitly out of scope.

The frozen minimality separation was

$$\delta_H = \min \left(\alpha(H), -\max_{G \subset H} \alpha(G) \right),$$

with primary threshold 0.02 s^{-1} . A positive alpha alone was not accepted as physicality evidence.

2 Discovery audit and frozen selection

The authoritative prior file contained 256 portfolios x 24 holdout conditions, or 6144 portfolio-condition evaluations. Its frozen summary reported 42 minimal H0 cases and 47 minimal H0.05 cases. This narrow campaign did not rerun the census.

The controls were frozen by same cardinality, then MW, MVA, remaining inertia, and lexicographic order. No selection was changed after observing results.

Table 1: Frozen candidate selection.

ID	Candidate	Condition	Matched control	Distance
H1	35;36	C12	32;36	0.0013860
H2	37;38	C12	36;38	0.0148840
H3	30;32;33	C12	30;33;35	0.0014786

3 Gate A - numerical and DAE physicality

The audit covered 17 unique portfolios at C12 and nominal with tolerances $1e-8$, $1e-10$, and $1e-12$. It used the native reduction, an independent dense spectrum, a descriptor/generalized check where exposed, central finite-difference diagnostics, conditioning, singular values, and participation. Within each fixed solved path, dense and descriptor spectra agreed with the native result. The equilibrium path did not.

Table 2: Default census initialization path at C12.

ID	alpha	frequency (Hz)	mode label	residual
H1	+10.1875267	0.000000	electromechanical/control	3.54e-12
H2	+22.5280341	0.549890	converter-control oscillatory	7.04e-13
H3	+21.1535104	0.000000	converter-control oscillatory	7.04e-13

Table 3: Strict Gate A path at tolerance $1e-10$.

ID	alpha	frequency (Hz)	interpretation
H1	-0.0979116	0.067999	stable on this path
H2	+11.8934740	0.000000	positive but path-sensitive
H3	-0.1088351	0.074147	stable on this path

At the default path, the immediate-predecessor separations for H1, H2, and H3 were 0.0971024, 0.0975787, and 0.0978716 s^{-1} . Those values are not physical minimality evidence once the candidate branch is shown to be path-sensitive. Large reduced-Jacobian conditioning, near-zero singular values, and PowerDynamics vertex-batch warnings are consistent with a numerical/DAE branch concern, but do not select a physical mechanism.

Gate A: FAIL_UNRESOLVED.

4 Gate B - common nonlinear TDS

The preregistered disturbance was bus 39 with a +1% P/Q constant-power-factor pulse from 1.0 to 1.1 s, restoration, and integration to 20 s with common Rodas5P settings. The run contained 26 rows. A `maxiters=100000` guard was added only to prevent unbounded runtime; it did not change the disturbance, tolerances, observables, or acceptance rule. A max-iteration return is a TDS failure.

All three selected C12 candidates returned `MaxIters`. Immediate predecessors and controls completed with negative estimated rates, roughly -0.207 to -0.258 s^{-1} . The failed candidate integrations showed large state/Jacobian excursions before the solver stopped. This pattern is not nonlinear physical confirmation.

Gate B: FAIL.

5 Gate C - composition

The composition gate was conditional on physical blocker confirmation. Because Gates A and B did not confirm a blocker, the gate was not entered. Raw default-path candidate/control comparisons are retained in

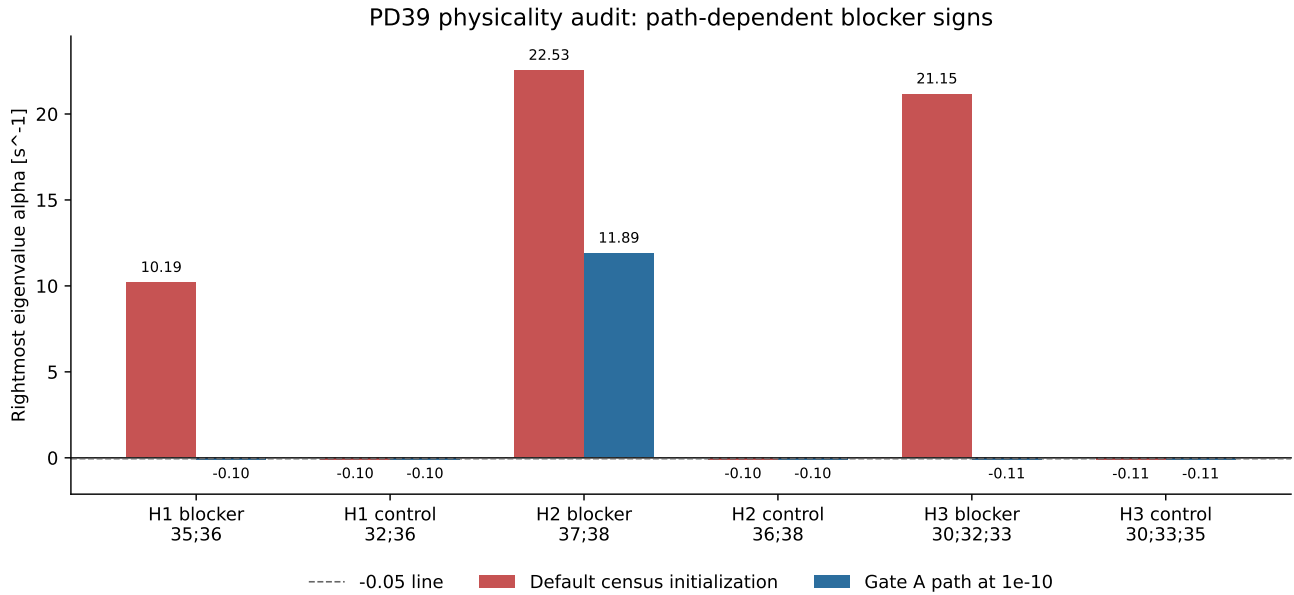


Figure 1: Default versus strict audit path for candidates and controls.

results/PD39_BLOCKER_COMPOSITION_TEST.csv; they do not constitute a composition claim. In particular, a large H2 default-path contrast cannot be promoted while the equilibrium branch is unresolved.

Gate C: NOT APPLICABLE; composition strong = false.

6 Penetration, composition, and mode mechanism

The prior holdout results show H0/H0.05 cases across multiple conditions rather than a universal V8 identity. In the selected C12 cases, the default labels are heterogeneous: H1 is electromechanical/control and H2/H3 are labeled converter-control by the default path. The tolerance/path changes and failed TDS integrations prevent a physical 7/8 -> 8/8 mechanism claim. No physical SG-to-GFL homotopy was run.

7 Exact network closure

The installed model/API does not expose the exact device-port objects, local admittance partition, and dimensioned K, D, and Q objects required for the requested contextual-return identity. PD contextual return is therefore **BLOCKED**. TX4 contextual-return validation was not run after the physicality STOP. No proxy, determinant identity, or Q -> -1 boundary was constructed.

Closure: BLOCKED.

8 Compatibility atlas and conditional theory

The PLL/filter two-coordinate atlas, 11 x 11 grid, adaptive boundaries, κ_0 , $\kappa_{0.05}$, mode-MAC continuation, contextual-return theorem validation, and local remediation were not run. A compatibility atlas on path-sensitive equilibria and failed TDS cases would overstate the evidence.

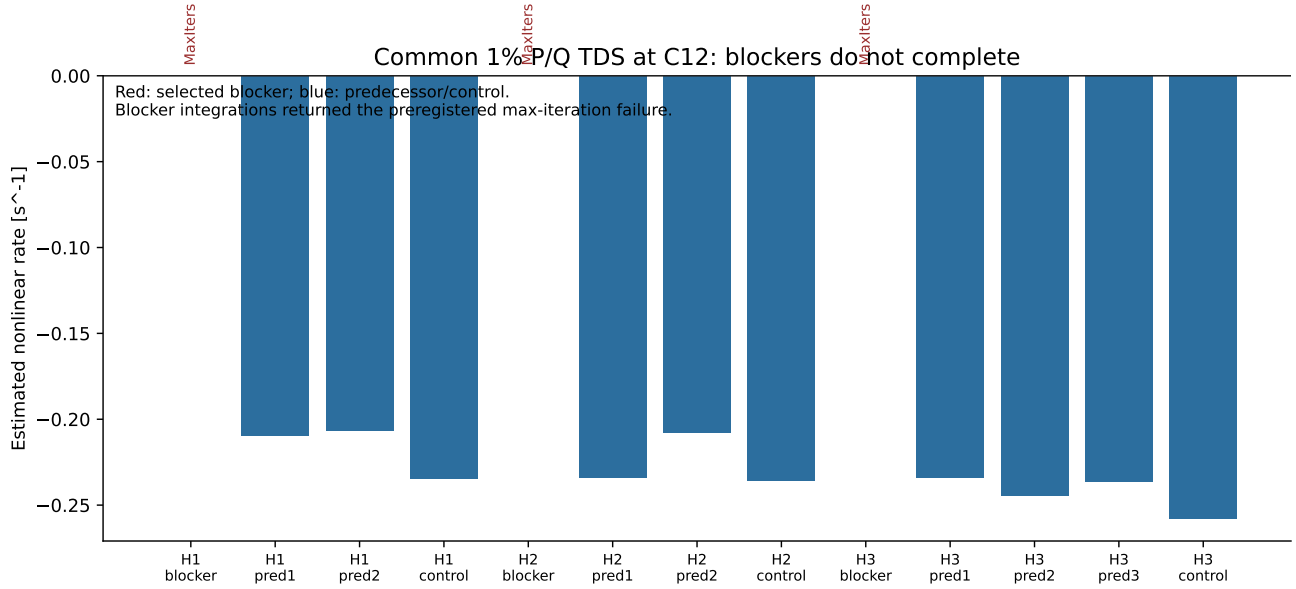


Figure 2: C12 TDS completion pattern. Red candidate bars are absent because the solver returned MaxIters; blue neighbors completed and decayed.

9 Negative results and forbidden claims

This branch does not support a physical minimal H0/H0.05 blocker, a physical 7/8 $\rightarrow$ 8/8 mode transition, an aggregate composition mechanism, compatibility boundaries, contextual-return closure, or any controller/line/topology repair. It also does not revive hidden-radius, generic weak-node/link, graph Laplacian/Fiedler, cycle/holonomy, Shapley, or broad co-design novelty.

The strongest defensible positive statement is narrower: the stock PD39 workflow produces low-order C12 instability candidates whose default-path signs are not reproducible under the frozen audit and whose common-disturbance trajectories can terminate in solver iteration failure while neighbors decay. This is a validation warning requiring model/equilibrium-path repair.

10 Practical interpretation and final decision

For SG-to-IBR planning, the immediate lesson is a quality-control rule: do not promote a portfolio to a physical blocker from one initialization and one rightmost eigenvalue. Require branch reproducibility, independent spectrum agreement, conditioning diagnostics, and a common nonlinear disturbance that completes.

The campaign does not justify redesigning the IAS poster around a physical blocker or atlas. It is not Transactions-ready as a new mechanism result.

FINAL CASE: C.

Reproducibility files

The final repository retains the preregistration, deviations, selection table, Gate A tables, default-path audit, Gate B raw traces, STOP figures, final machine-readable claims, and two ZIP bundles. The compact bundle is `deliverables/PD39_BLOCKER_ATLAS_CHATGPT_UPLOAD.zip`.

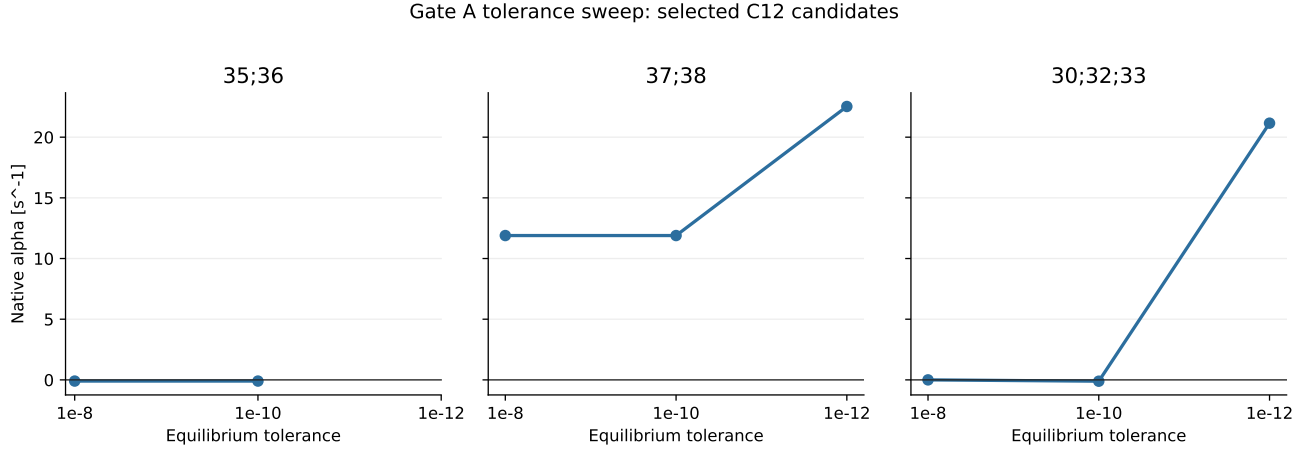


Figure 3: Frozen tolerance sweep for selected C12 candidates.

Table 4: Final status matrix.

Item	Status
Numerical audit	FAIL_UNRESOLVED
Selected-blocker TDS	FAIL: 3 C12 MaxIters
Prior H0 count over 24 holdouts	42
Prior H0.05 count over 24 holdouts	47
Penetration/composition/mixed verdict	no physical blocker
Mode mechanism	unresolved, path-dependent labels
Closure	BLOCKED
Atlas	NOT RUN_STOP
IAS poster redesign	NO
Transactions-ready	NO